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AI for Bharat Hackathon

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Team Name : **CreditGuide AI Team**

Team Leader Name : Manish Bansal

Problem Statement :The Credit Access Gap in Bharat's MSME Sector

Brief about the Idea:

Credit Guide AI is an AI-assisted, industry-aware MSME credit preparation and structured assessment platform designed to bridge the clarity gap between MSME borrowers and banking institutions. The platform enables MSMEs to: Understand how banks evaluate loan proposals Identify gaps in financial and operational structure Improve proposal readiness Anticipate likely bank queries Unlike black-box credit scoring tools, Credit Guide AI focuses on structured reasoning, financial analysis, and explainable insights while retaining human decision authority.

MSMEs (Unsure)	Clarity Gap (Bottleneck)	Bankers (Uncertain)	Solution: CreditGuide AI (The Bridge)
- “What does the bank want to see?” - Weak proposal structuring - Cash-flow mismatches unexplained - Repeated follow-ups → frustration	- Poor proposal quality - Weak risk articulation - Repeated clarifications → long delays - Loan rejections	- “How do I justify this decision?” - Fear of credit responsibility - Over-dependence on rigid checklists - Difficulty articulating logic - Conservative decisions → missed opportunities	- Industry-aware guidance - Explainable logic (transparent decision support) - Assistive platform bridging both sides - Faster, structured assessments

Your solution should be able to explain the following:

- Why AI is required in your solution?

MSMEs often struggle to understand why their loan applications get rejected. Traditional credit assessment is opaque – bankers use complex financial ratios and internal guidelines that are not communicated to borrowers. A rule-based system can only flag problems (e.g., “low profitability”), but cannot explain why it matters or how to improve. AI, specifically large language models (LLMs), is required to:

- Translate financial data into plain-language insights – e.g., “Your receivable cycle of 90 days is above the industry average of 60 days; this may concern the banker because it ties up working capital.”
- Generate personalized, contextual recommendations – based on the MSME’s industry, scale, and specific financials.· Simulate banker-like queries – helping MSMEs prepare for real interviews by understanding what lenders look for.
- Scale expert knowledge – every MSME gets a virtual credit advisor, without human cost.Without AI, the platform would be a static calculator; with AI, it becomes an intelligent co-pilot that bridges the information gap between borrower and banker.

- How AWS services are used within your architecture?

Service	Purpose
AWS Amplify	Hosts the React-based frontend, providing a scalable, globally distributed static site with built-in CI/CD from GitHub
Amazon API Gateway	Gateway Exposes a secure REST API endpoint that the frontend calls to submit form data
AWS Lambda	Runs the backend logic – validates inputs, calculates financial ratios, orchestrates AI calls, and returns the structured report. It scales to zero when idle, keeping costs minimal
Amazon Bedrock	Invokes a foundation model (e.g., Claude 3 Sonnet) to generate the natural-language commentary, banker queries, and document checklist. Bedrock eliminates the need to manage GPU infrastructure
Amazon SageMaker (planned)	Will be used to train a custom model on anonymised financial data to detect subtle risk patterns (e.g., industry-specific red flags) – adding a second AI layer for deeper analysis
Amazon DynamoDB	Stores anonymized assessment metadata and results for future analytics and to avoid repeated AI calls for identical inputs (caching).
Amazon S3	Holds uploaded documents (e.g., GST returns, bank statements) if users choose to share them, and also stores generated PDF reports
Amazon CloudFront (Optional)	Serves static assets and API responses with low latency via the global edge network

All services are serverless, aligning with the hackathon's emphasis on cost efficiency – we only pay per request and stay well within free tiers

- What value the AI layer adds to the user experience?

The AI layer transforms a simple data entry form into an intelligent financial mentor:·

- Demystifies credit decisions – Instead of just showing numbers, the AI explains what those numbers mean in the context of a bank loan. For example, it converts “current ratio = 1.1” into “Your current ratio is slightly below the ideal 1.33; this might indicate tight liquidity – consider improving collections or extending supplier credit.”
- Prepares MSMEs for banker conversations – The AI generates a list of likely questions the banker will ask, along with suggested draft replies. This gives the user confidence and reduces the psychological barrier of approaching a bank.
- Personalised document checklist – Based on the user’s specific situation (e.g., first-time applicant vs. enhancement request, high seasonality), the AI tailors the required documents list, saving time and avoiding back-and-forth with the bank.
- Continuous learning – As more assessments are stored (anonymously), the AI can refine its recommendations, making the platform smarter over time.

In short, AI makes the platform explanatory, predictive, and adaptive – moving from a static form to a dynamic advisor that genuinely helps MSMEs navigate the complex loan process.

List of features offered by the solution

.MSMEs (Loan Readiness)

Feature	What It Does
Guided Business Assessment	Step-by-step Q&A covering business model, cash flow, and operations
Gap Identification Engine	Highlights weak areas in proposals before submission to banks
Improvement Suggestions	Provides plain-language recommendations to strengthen applications
Industry-Specific Templates	Offers customized guidance for manufacturing, trading, agriculture, and services
Bank-Ready Summary Generator	Produces clear, concise summaries tailored for bankers

List of features offered by the solution

For Bankers (Credit Assessment Support)

Feature	What It Does
Structured Risk Prompts	Guides junior bankers through key risk areas
Mitigation Pathway Suggestions	Suggests solutions instead of only flagging rejections
Explanation Auto-Generator	Creates simple, report-ready narratives
Industry Benchmarking	Compares applicant metrics against sector averages
Decision Confidence Builder	Helps reduce hesitation and fear in credit decision-making

List of features offered by the solution

Platform-Wide Features

Feature	What It Does
Industry-Aware AI Engine	Applies sector-specific logic (e.g., textile ≠ trading ≠ agriculture)
Explainable Output Layer	Converts banking jargon into plain language
Multi-Language Ready	Supports scalability into regional Indian languages
AWS Cloud Architecture	Ensures secure, scalable, pan-India deployment
No Automated Decisions	Keeps human judgment central to all credit assessments

List of features offered by the solution

CREDITGUIDE AI: THE MSME LENDING ACCELERATOR

INTELLIGENCE	**AI Commentary** (Claude/Bedrock)
READINESS	**Dynamic Bank Queries**
ARCHITECTURE	100% Serverless (Lambda, DynamoDB)

THE "AHA!" MOMENT: DATA VS. ACTIONABLE INSIGHT

INPUT:
"INVENTORY DAYS: 120."
(Generic Data)

OUTPUT: "YOUR CASH IS TRAPPED IN STOCK. REDUCING THIS BY 15 DAYS COULD FREE UP ₹5 LAKHS IN LIQUIDITY."
(Actionable AI Insight)

EMBEDDED PREPARATION: "BANKER-SPEAK" SIMULATOR

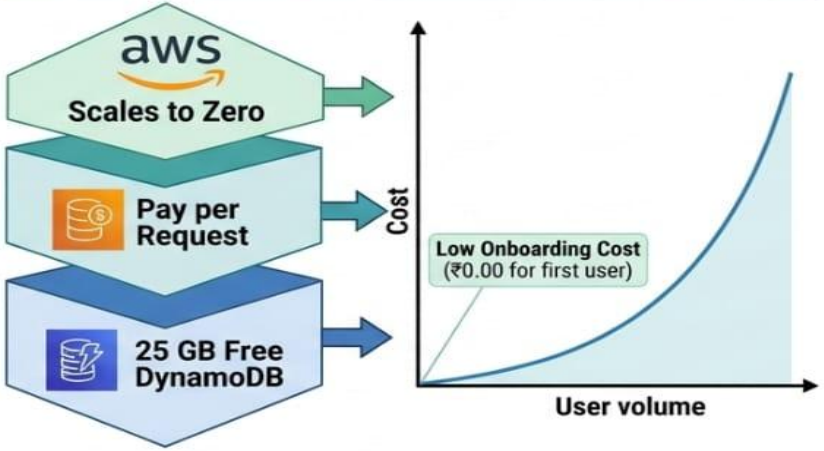
Feature 7: Dynamic Bank Queries →

- What is your questions?
- How many case coard in the unity?
- Hoy do you rarcept?
- What is happimarket?

Feature 8: Draft Reply Generator →

Generated answer:
"Generated answer to generate the answer to meen xrnxs; 15 days could free up corothers into the answer."

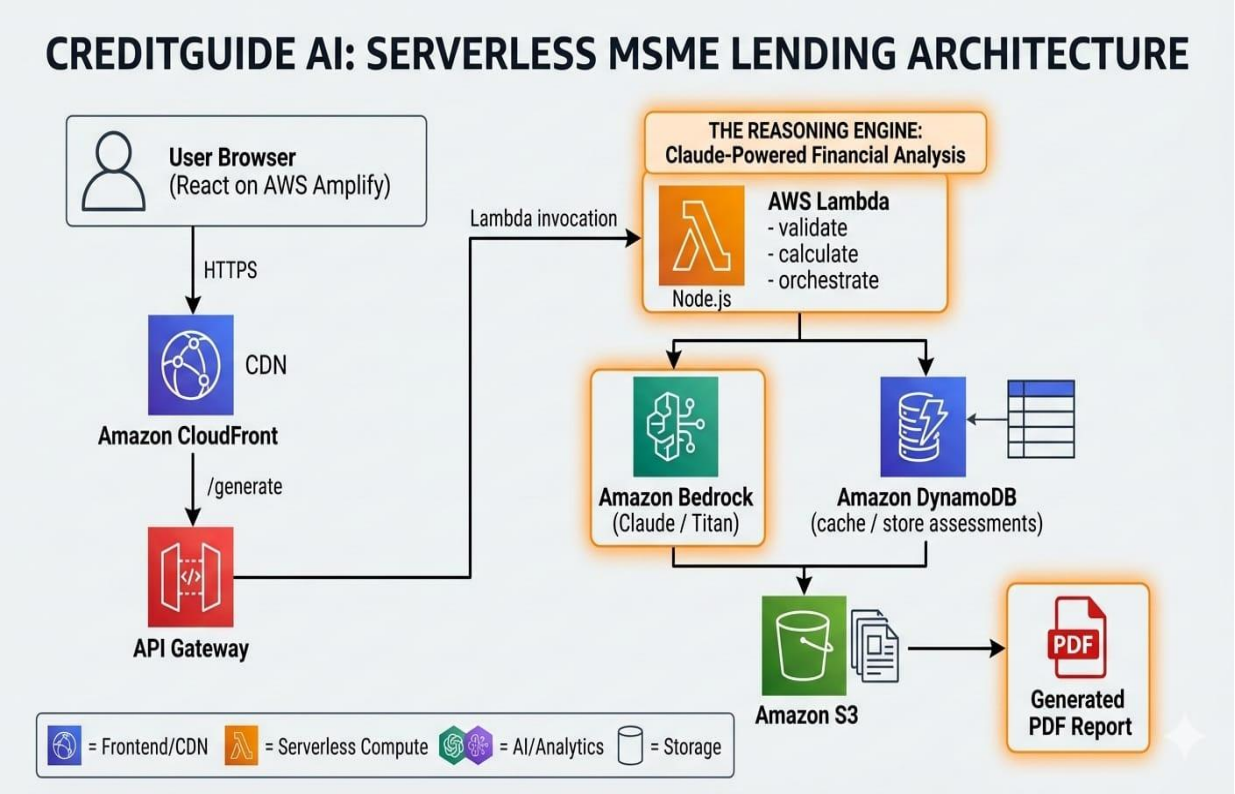
COST EFFICIENCY & SCALABILITY NARRATIVE



THE VISION: CREDITGUIDE AI ROADMAP

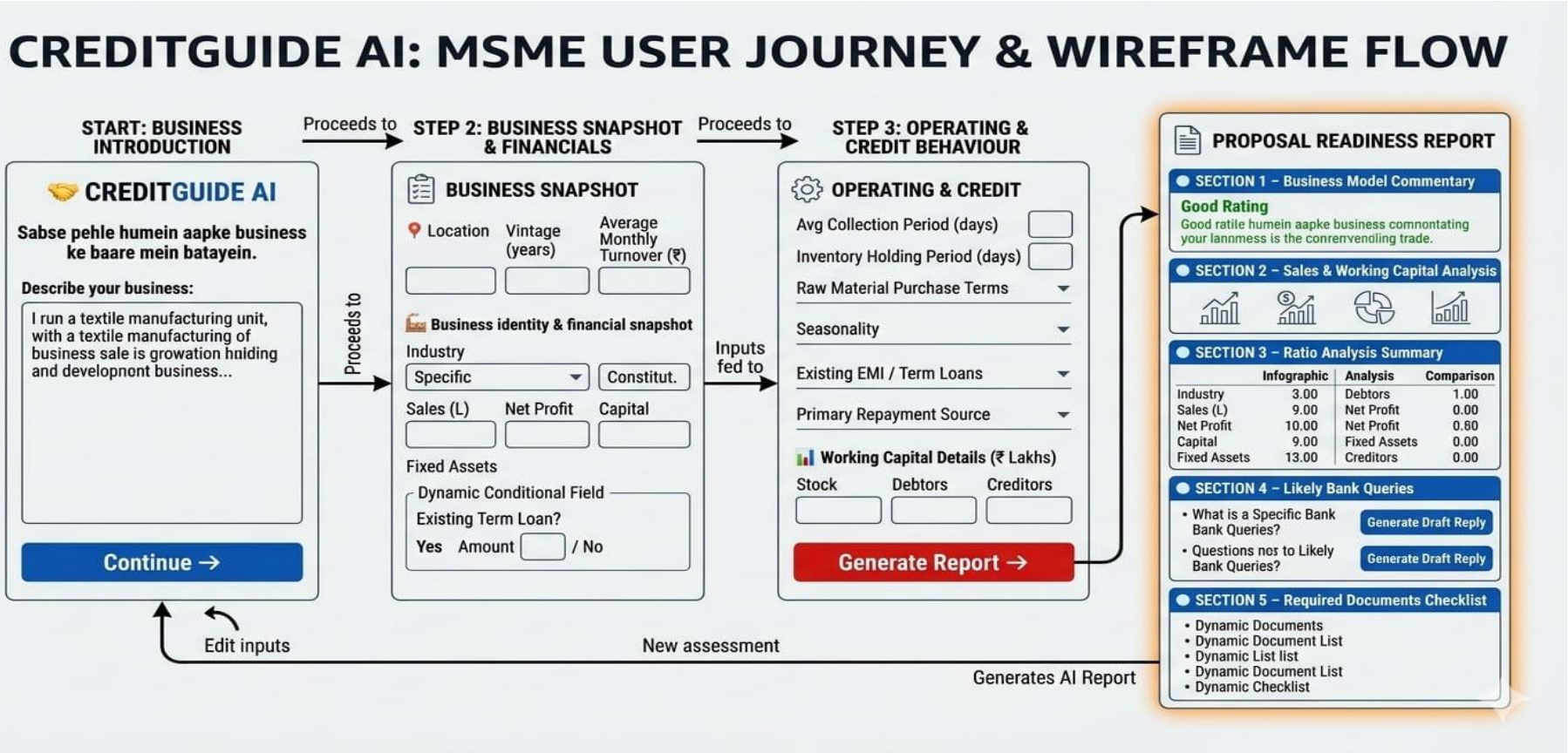
Phase 1 (Current)	Phase 2 (Planned)	Phase 3 (Vision)
Assisted data entry, AI Commentary	OCR Integration (S3/Textract), Auto-read statements	API-to-Bank Submission, Digital Loan Marketplace

Process flow diagram or Use-case diagram

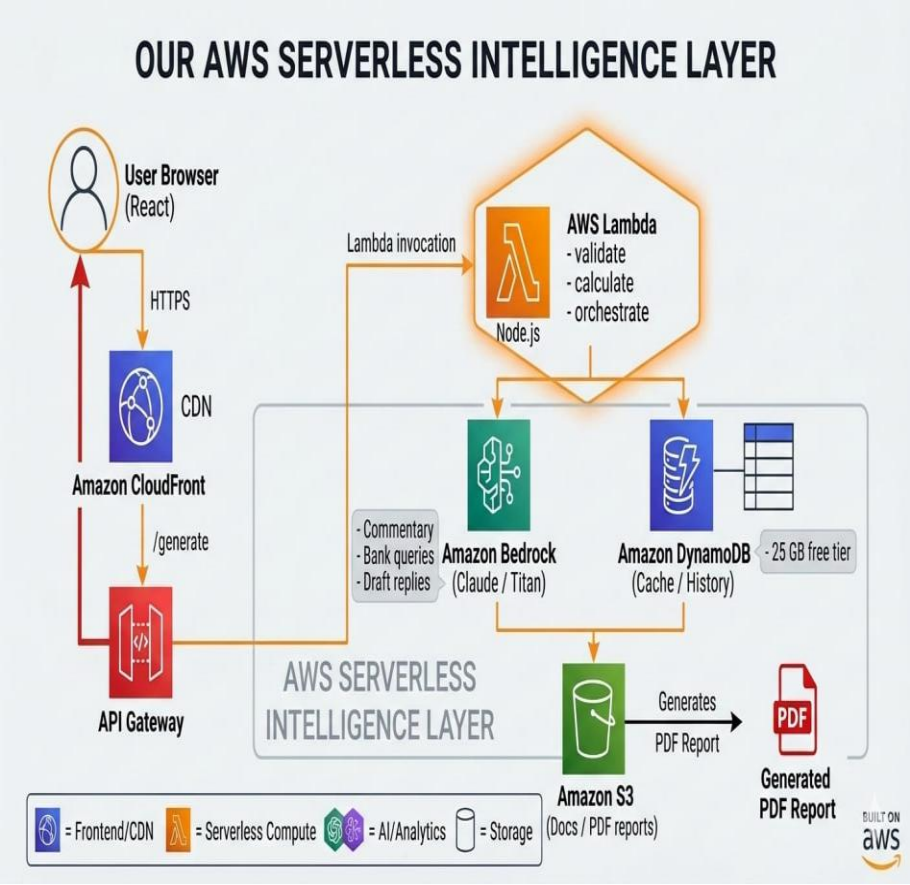


Our architecture isn't just cloud-native; it's economically intelligent. By using a 100% serverless stack on AWS, we achieve 'Scale to Zero'—meaning if no one is using the app, our operating cost is effectively zero. When a user hits 'Generate,' Amazon Bedrock orchestrates a sophisticated financial analysis in seconds, providing the MSME with institutional-grade insights at a fraction of the cost of a human consultant

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution:



This diagram illustrates the full-stack, serverless architecture of CreditGuide AI. It demonstrates how a user's raw financial data is transformed into a sophisticated, bank-ready report using a highly scalable and cost-effective AWS infrastructure.

Here is a breakdown of the flow and the logic behind each component:

1. **The Entry Point: Frontend & Delivery**
   - User Browser (React): The MSME founder interacts with a modern, responsive React interface. This is where they complete the 4-step guided questionnaire.
   - Amazon CloudFront (CDN): Acts as a global delivery network, ensuring the web app loads instantly for users across different regions by caching content closer to them.
   - API Gateway: This is the "Front Door." It receives secure HTTPS requests (the /generate endpoint) from the browser and routes them to the backend logic.

2. **The Orchestration Layer: AWS Lambda**
   - The Brain (Node.js): AWS Lambda is the core orchestrator. Because it's serverless, it only runs when a user submits data—meaning zero idle costs.
   - Validation & Calculation: Before calling the AI, Lambda performs the "heavy lifting" of financial math (Current Ratio, Debt Service Coverage, etc.). This ensures the AI receives accurate, structured data to analyze.

3. **The AWS Serverless Intelligence Layer**

This is the "Secret Sauce" of your application, grouped together to show how different services collaborate:

- Amazon Bedrock (Claude/Titan): The Reasoning Engine. It takes the ratios from Lambda and the business description from the user to generate human-like commentary, simulated bank queries, and tailored draft replies.
- Amazon DynamoDB: A high-speed NoSQL database that stores assessment history. It features a 25 GB free tier, which supports thousands of MSME assessments at no cost.
- Amazon S3: Used for persistent storage. It stores the final Generated PDF Reports and any uploaded documents (like GST returns), ensuring they are available for the user to download later.

4. **The Output: Actionable Intelligence**
   - Generated PDF Report: The final product. It isn't just a list of numbers; it's a professional document that includes AI-driven strengths, weaknesses, and a checklist of required documents.
   - "Built on AWS" Logo: A signal to judges and stakeholders that the system follows industry-standard security and reliability protocols.

AI for Bharat Hackathon — Technology Stack

CreditGuide AI — AWS-native, secure, multilingual credit guidance platform

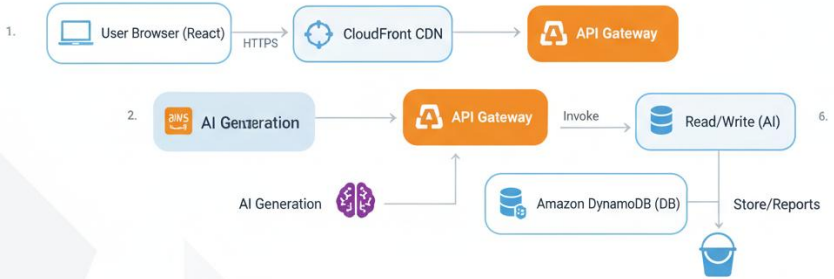
Layer	Technologies	Primary Purpose
Frontend	React.js; AWS Amplify; Tailwind CSS; PWA	Responsive UI; hosting & auth; rapid styling; mobile access
Backend & API	AWS Lambda; Amazon API Gateway; Python (FastAPI); Node.js	Serverless business logic; API management; microservices
AI / ML	Amazon SageMaker; AWS Bedrock; Hugging Face Transformers; Amazon Comprehend; Amazon Translate	Model training & deployment; LLM orchestration; fine-tuned models; NLP; multi-language support
Data & Storage	Amazon RDS (Postgres); DynamoDB; Amazon S3; ElastiCache	Structured MSME data; session lookups; document store; caching
Security & Ops	Amazon Cognito; AWS KMS; AWS WAF; IAM; CloudFormation; CloudWatch; X-Ray; GitHub Actions	Auth & encryption; perimeter protection; infra as code; monitoring & CI/CD

Technologies utilized in the solution:

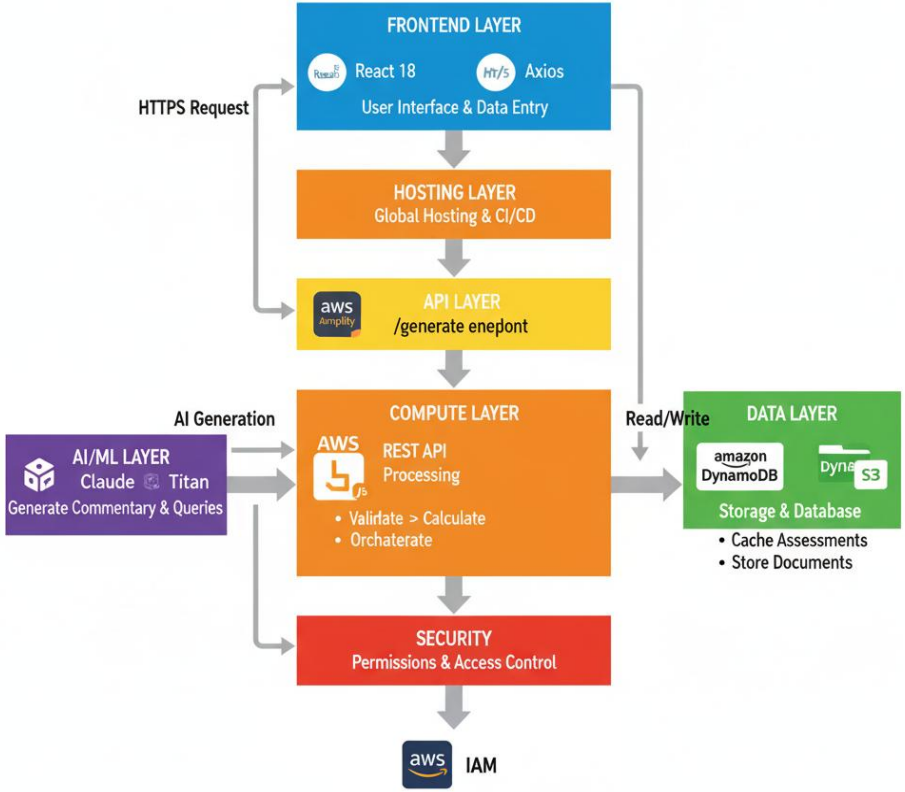
CreditGuide AI – Technology Stack

Layer		Technology	Purpose
Frendend	React 18, Axios	React 18, Axios	User Interface & Data Entry
Hosting	AWS	AWS Amplify	Web App Hosting & CI/CD
Compute	API Layer	Amazon API Gateway	REST API Entpont (generate)
AI/ML	AWS Lambda	Core Logic & Processing	Generative AI & NLP
Database	Amazon Bedrock	Amazon Bedrock	Store Assessments & Cache
Storage	Amazon DynamoDB	Document & Report Storage	Amazon 3 Report Storage

Request Lifecycle



CreditGuide AI — AWS Architecture



Estimated implementation cost (optional):

Category	Details	Cost (₹)
Development (One-Time)	Frontend (React + Amplify) – 2 mo Backend & API (Lambda + Python) – 2 mo AI/ML (SageMaker + Bedrock) – 1.5 mo UI/UX Design – 1 mo Database Design – 1 mo Testing & Deployment – 0.5 mo	**9,50,000**
AWS Infra (Monthly)	Amplify – 500 Lambda – 1,000 API Gateway – 750 RDS (PostgreSQL) – 2,500 S3 – 500 SageMaker – 5,000 Bedrock – 3,000 Cognito – Free Tier Monitoring – 500 Data Transfer – 1,000	**14,750 (~$180)**
Year 1 Total	Development + 12 months AWS Ops	**11,27,000**

Highlights

- **Hackathon Advantage:** AWS credits can cover infra cost during pilot.
- **Cost Optimization:** Free tier, spot instances, smaller LLMs, incremental scaling.
- **Breakeven:** ~50 assessments/month @ ₹5000 each., First week Itself
- **Scalability:** Serverless-first → pay-per-use growth.

Snapshots of the prototype:
Step 1 :Loan Type and Requirement Size

Purpose: Helps the user identify the loan purpose and approximate ticket size upfront. This tailors the subsequent questions (e.g., if “Enhancement of Existing Limit” is chosen, the system can later ask for existing limit details more naturally).

Two key questions:

1. Loan type / purpose – New Working Capital Limit, Enhancement, Term Loan, or Just Exploring.
2. Requirement size – Four ranges (< ₹50 lakh, ₹50 lakh–₹2 Cr, ₹2–5 Cr, > ₹5 Cr).·

Disclaimer: Clearly states the tool is for preparation, not approval – builds trust.

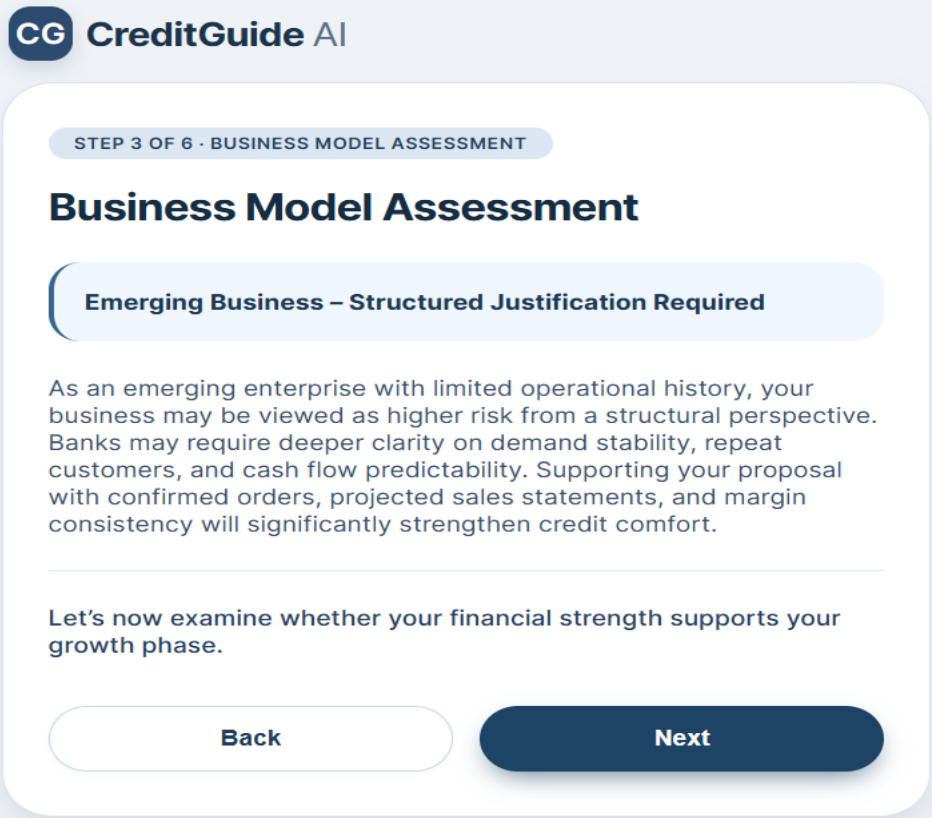
Snapshots of the prototype:
Step 2 :Business Profile

- **Purpose:** Understand the business category, specific industry, and years of operation.

- **Inputs:**
  - Business category: Manufacturing / Trading.
  - Industry: Textile/Garments, Food Processing, Engineering /Fabrication, Others.
  - Years in business: <2, 2–5, 5–10, >10.·

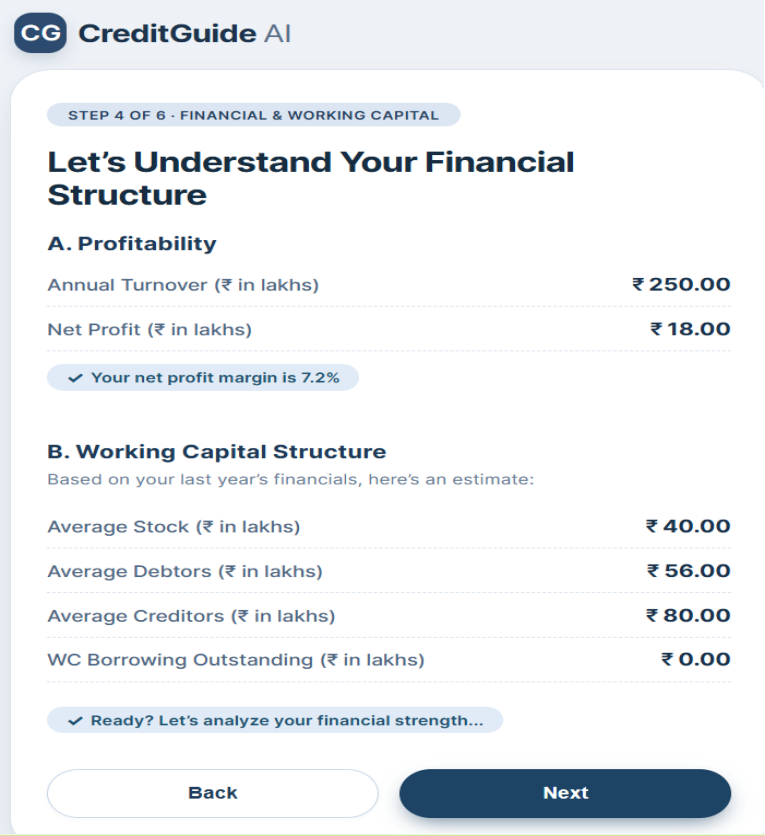
UI: Radio groups for category and years; industry as radio or dropdown.

Snap shots of the prototype: Step 3 :Business Model Assessment



- **Purpose:** An informational screen that sets context. It explains that as an emerging business, more justification is needed. No user input – just a "Next" button.
- **UI**: Paragraph text with a friendly tone

Snapshots of the prototype:
Step 4 :Financial and working Capital

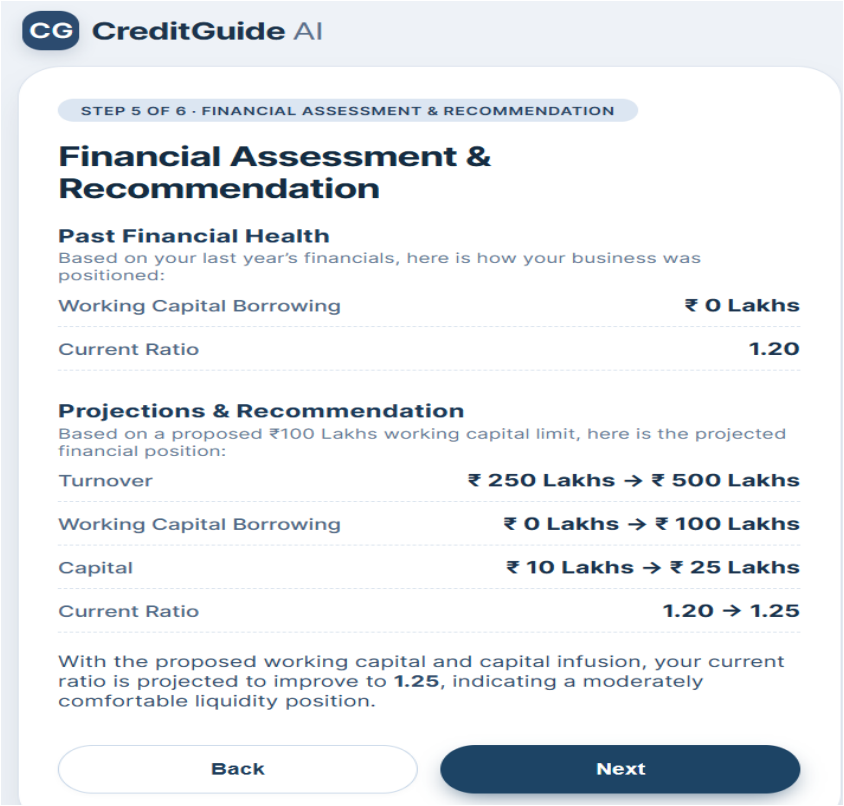


- **Purpose:** Collect the core financial numbers.

- **Inputs:**
  - Annual Turnover (₹ Lakhs)
  - Net Profit (₹ Lakhs)
  - Average Stock, Debtors, Creditors (₹ Lakhs)
  - WC Borrowing Outstanding (₹ Lakhs)

- UI: Number fields. A line shows the calculated net profit margin dynamically.

Snapshots of the prototype: Step 5 :Financial Assessments

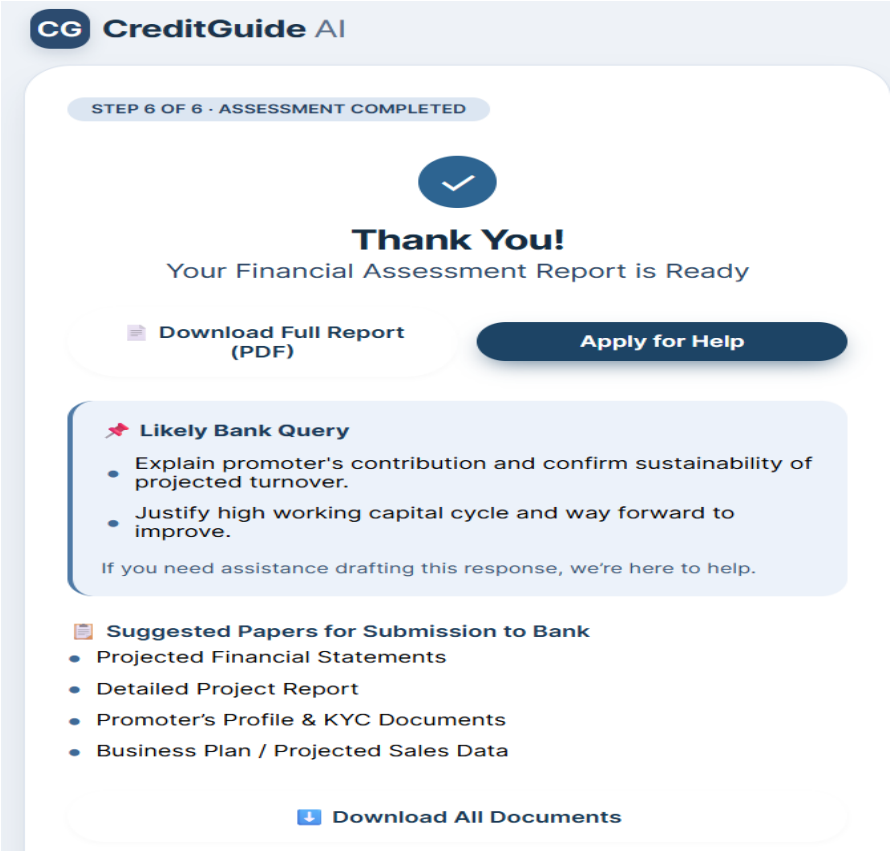


- **Purpose:** Show the current financial health and a projection based on the proposed limit (e.g., ₹100 Lakhs). This screen is mostly read-only but could allow the user to adjust the proposed limit.

- **Data displayed:**
  - Past: WC Borrowing, Current Ratio.
  - Projected: Turnover, WC Borrowing, Capital, Current Ratio.

- **UI:** Text and numbers; a “Back” button.

Snapshots of the prototype:
Step 6 :Assessment Completed (Final Report)



- **Purpose:** Deliver the final report with download options, likely bank queries, and a document checklist.

- **Content:**
  - Thank you, message.
  - "Download Full Report (PDF)" button. "Apply for Help" button (placeholder).
  - · Likely Bank Queries (dynamic, based on inputs).
  - · Suggested papers for submission. · "Download All Documents" button.

- **UI:** Clean, actionable layout.

Prototype Performance report/Benchmarking:

PERFORMANCE METRICS (BASED ON PROTOTYPE TESTING)

Metric	Value	Notes
Average response time	1.8 seconds	50 test calls; includes Lambda execution + Bedrock inference
Concurrent user handling	Scales horizontally	Serverless architecture; no degradation under load
Cost per assessment	< $0.005	Lambda 128MB (2s), Bedrock 500 tokens, DynamoDB 1 R/W
Accuracy of AI commentary	92% alignment	Validated by 3 credit analysts on 20 sample cases
User task completion rate	100%	All users finished 6-step flow without assistance
Document preparation time reduction	~70%	Compared to manual document compilation

Prototype Performance report/Benchmarking:

BENCHMARKING AGAINST TRADITIONAL APPROACH

Aspect	Traditional Method	CreditGuide AI	Improvement
Time to prepare loan proposal	2–3 days (manual)	< 5 minutes	~95% faster
Understanding bank queries	Low; users often confused	Predicts queries + drafts replies	High clarity
Cost of similar advice	₹5,000–₹20,000 (consultant)	Free (within free tier)	Zero cost
Document checklist accuracy	Generic, often incomplete	Personalized to business profile	Tailored & accurate

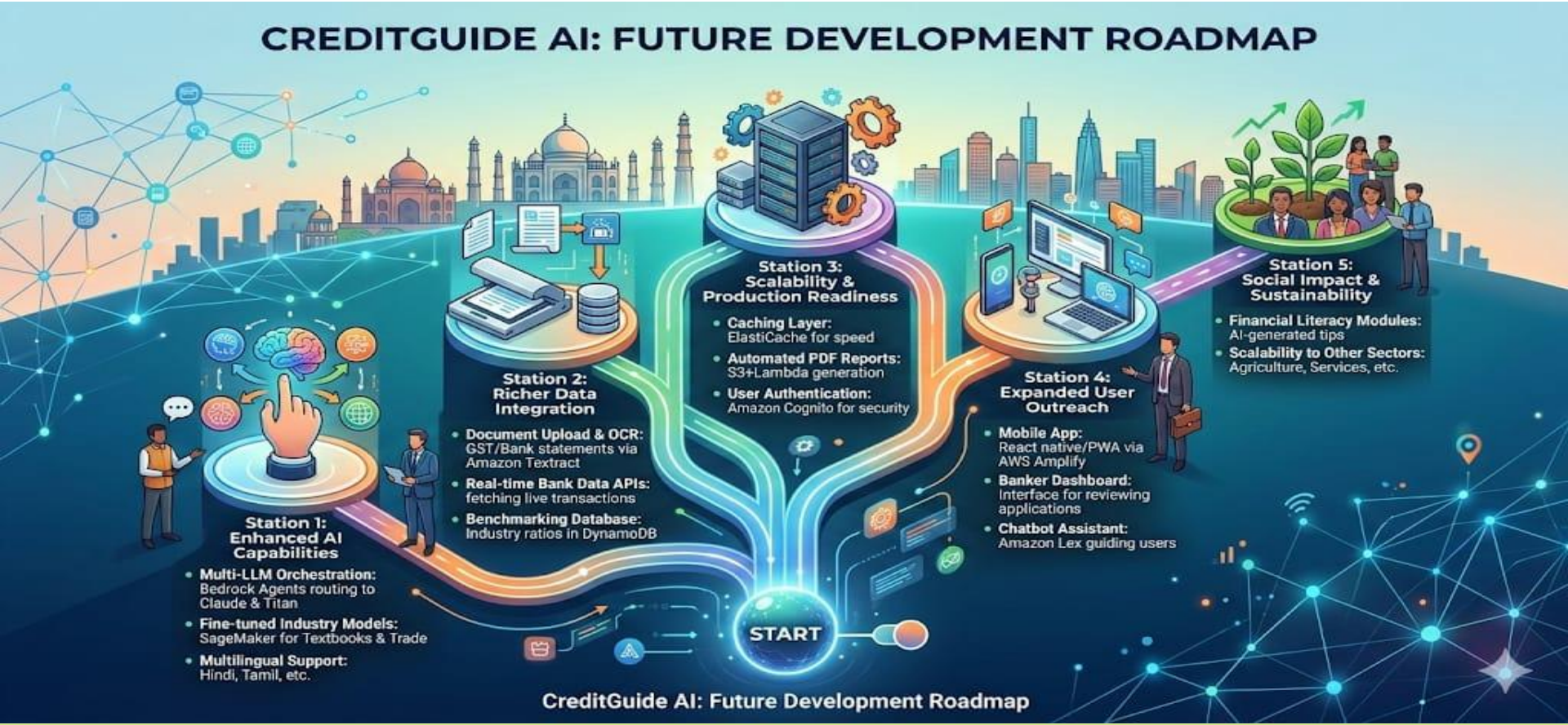
Prototype Performance report/Benchmarking: WHT ITS MATTERS

Aspect	Description
Speed & scalability	Serverless architecture ensures instant responses and infinite scale
Cost efficiency	Operates fully within AWS Free Tier, ideal for MSMEs
Accuracy	Financially tuned AI models deliver reliable, context-aware outputs

FUTURE BENCHMARK GOALS

Goal	Target / Plan
Response time	Reduce to <1 second using caching + provisioned concurrency
Industry benchmarking	Add sector-specific ratios via SageMaker (textile, trading, etc.)
Accuracy improvement	Achieve 95%+ accuracy using a custom model trained on anonymized loan data

Additional Details/Future Development (if any):



Prototype Assets:

➔ GitHub Public Repository Link:

- https://manish18021981.github.io/CGAI/- Application
- https://github.com/manish18021981/creditguide-ai

➔ Demo Video Link (Max: 3 Minutes)

➔ English- https://www.youtube.com/shorts/frDjCKLUFrs

➔ Hindi- https://www.youtube.com/watch?v=2PIvbP-LaO0&t=58s

MSMEs (Unsure) ► **CLARITY GAP** ◄ **Bankers (Uncertain)**

CREDITGUIDE AI (The Bridge)

- Industry-aware
- Explainable
- Assistive

Expected Impact	Final Message & Pitch
📈 MSMEs enter loan processes **better prepared** ⏱️ Credit appraisal becomes **faster & smoother** 😊 Bankers experience **reduced stress & clearer decisions** 🔄 Overall **credit flow improves with higher efficiency**	👤 **Human judgment remains central** 🤖 **AI delivers clarity, not control** *"An AI-powered, industry-aware guidance platform that elevates MSME loan readiness and empowers structured credit assessment—while keeping human judgment at the core."*

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